

Differentially Private Prompts Warehouse

1 Introduction

Utilizing differential privacy mechanisms to perturb tokens in the prompts has become a promising method for privacy protection in large language models. A series of works [YDW⁺21, CMW⁺23, TCZ⁺25, WDL⁺25] perturb each (sensitive) token by perturbing it to another with a specially designed exponential mechanism. This requires huge computational overhead to protect one prompt given a privacy budget. Consider a prompts warehouse keeps many private prompts. There are users who ask for some prompts with special privacy budgets.

Strawman Solution 1. Compute the private prompts with given budget on-the-fly, which leads huge computational overhead and long response time.

Strawman Solution 2. Pre-compute several private prompts with default budget (i.e., $\varepsilon = 1, 2, 4, 8, \dots$) and return one of perturbed prompts whose budget smaller than given one. Since smaller budget leads to higher protection but lower accuracy, this method leads to poor utility.

We aim to design a mechanism achieving both performance and utility.

2 Background

2.1 Exponential Mechanism

For any $\varepsilon > 0$, a randomized exponential mechanism $\mathcal{R}_{\text{EM}} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies ε -DP if, for each input $x \in \mathcal{X}$, the mechanism outputs $y \in \mathcal{Y}$ with probability:

$$\Pr[\mathcal{R}_{\text{EM}}(x) = y] \propto \frac{\varepsilon}{2\Delta} \cdot u(x, y),$$

where $u(x, y)$ is a user-defined utility function and $\Delta := \max_{x \sim x', y} |u(x, y) - u(x', y)|$ is the global sensitivity.

2.2 Prompt Perturbation via DP

The naive DP mechanism works as follow

1. Divide the prompt x into tokens $(t_1, t_2, \dots, t_n)$, where $t_i \in \mathcal{V}$ and $\mathcal{V}$ is the vocabulary;
2. For each (sensitive, which is defined by a subset $\mathcal{V}_S \subseteq \mathcal{V}$) token t_i , compute its token embedding distance between all tokens $t \in \mathcal{V}$, denoted as $d(t_i, t)$;
3. Compute the utility function $u(t_i, t) \leftarrow f(d(t_i, t))$;
4. Use EM mechanism to sample a perturbed token and output it.

2.2.1 SanText

<https://github.com/xiangyue9607/SanText>

SanText adopts the geometry of differential privacy (or, metric local differential privacy): Given $\varepsilon > 0$ and a distance metric $d : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$, for any $x \sim x'$, the mechanism satisfies ε -GP (or $\varepsilon \cdot d(x, x')$ -DP) if

$$\Pr[\mathcal{M}(x) = y] \leq e^{\varepsilon \cdot d(x, x')} \cdot \Pr[\mathcal{M}(x') = y].$$

SanText simply applies EM mechanism with $u(t_i, t) \leftarrow d(t_i, t)$, while setting $\Delta = 1$. It thus satisfies ε -GP, which is equivalent to $\varepsilon \cdot d_{\max}$ -DP, where $d_{\max}$ is the maximum distance in $\mathcal{V}$.

SanText⁺ divides the input to sensitive/non-sensitive sets $\mathcal{V}_S, \mathcal{V}_N$ and the output to protected/unprotected sets $\mathcal{V}_P, \mathcal{V}_U$

2.2.2 CusText

<https://github.com/sai4july/CusText>

2.2.3 InferDPT

<https://github.com/mengtong0110/InferDPT>

2.2.4 Cape

3 Idea

3.1 Findings

Different from the general EM mechanism, DP prompt perturbation has an interesting phenomenon:

1. The neighboring instance defines all tokens in the vocabulary.
2. The utility function (always based on token embedding distance) has commutative property.

Thus, we have $|u(t_i, t) - u(t_i, t')| \leq \Delta$, for all $t, t', t_i \in \mathcal{V}$. For any t and t' ,

$$\frac{\Pr[\mathcal{R}(t_i) = t]}{\Pr[\mathcal{R}(t_i) = t']} = \frac{\exp(\frac{\varepsilon}{2} \cdot \frac{u(t_i, t)}{|\Delta|})}{\exp(\frac{\varepsilon}{2} \cdot \frac{u(t_i, t')}{|\Delta|})} \leq e^{\varepsilon/2}.$$

Also

$$\sum_{t \in \mathcal{V}} \Pr[\mathcal{R}(t_i) = t] = 1.$$

We can derive:

$$\begin{aligned} \frac{e^{-\varepsilon/2}}{e^{-\varepsilon/2} + |\mathcal{V}| - 1} &\leq \Pr[\mathcal{R}(t_i) = t] \leq \frac{e^{\varepsilon/2}}{e^{\varepsilon/2} + |\mathcal{V}| - 1}, \\ \frac{e^{-\varepsilon/2}}{|\mathcal{V}|} &< \Pr[\mathcal{R}(t_i) = t] < \frac{e^{\varepsilon/2}}{|\mathcal{V}|}. \end{aligned}$$

3.2 Sample Amplification

Consider a prompt $(t_1, t_2, \dots, t_n)$ and its perturbed prompt $(pt_1, pt_2, \dots, pt_n)$ with privacy budget ε . Consider the following sample mechanism $\mathcal{S}$: for each (sensitive) token t_i , output pt_i with probability p ; or random sample one token from $\mathcal{V}$ otherwise. With a carefully decided p , the mechanism can achieve ε' -DP ($\varepsilon' < \varepsilon$).

Proof. Since the perturbed prompt satisfies ε -DP, for any $t_i, t'_i \in \mathcal{V}$, we have,

$$\frac{\Pr[\mathcal{R}(t_i) = pt_i]}{\Pr[\mathcal{R}(t'_i) = pt_i]} \leq e^{\varepsilon}.$$

The refined probability is:

$$\Pr[\mathcal{S} \circ \mathcal{R}(t_i) = pt_i] = p \cdot \Pr[\mathcal{R}(t_i) = pt_i] + (1-p) \cdot \frac{1}{|\mathcal{V}|}.$$

Let $q = |\mathcal{V}| \cdot \Pr[\mathcal{R}(t'_i) = pt_i] > e^{-\varepsilon/2}$, then

$$\frac{\Pr[\mathcal{S} \circ \mathcal{R}(t_i) = pt_i]}{\Pr[\mathcal{S} \circ \mathcal{R}(t'_i) = pt_i]} = \frac{p \cdot |\mathcal{V}| \cdot \Pr[\mathcal{R}(t_i) = pt_i] + (1-p)}{p \cdot |\mathcal{V}| \cdot \Pr[\mathcal{R}(t'_i) = pt_i] + (1-p)} \leq \frac{p \cdot e^{\varepsilon} q + (1-p)}{p \cdot q + (1-p)}. \quad (1)$$

Solve for (1) $\leq e^{\varepsilon'}$, we obtain

$$p \leq \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon} - e^{\varepsilon'})q + e^{\varepsilon'} - 1} < \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon} - e^{\varepsilon'})e^{-\varepsilon/2} + e^{\varepsilon'} - 1} = \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon/2} - 1)(e^{\varepsilon' - \varepsilon/2} + 1)}$$

Thus, by setting $p = \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon/2} - 1)(e^{\varepsilon' - \varepsilon/2} + 1)}$, the new mechanism satisfies ε' -DP. $\square$

Remarks. Some interesting results of ε' and p :

1. $\varepsilon' = 0$, then $p = 0$, which is a uniformly random sample;
2. $\varepsilon' = \varepsilon/2$, then $p = 1/2$.

3.3 Generalization

If we have two perturbed prompt: $(a_1, a_2, \dots, a_n)$ with privacy budget ε_1 ; and $(b_1, b_2, \dots, b_n)$ with privacy budget ε_2 . Consider the following sample mechanism $\mathcal{S}_2$: for each (sensitive) token t_i , output a_i with probability p ; or b_i otherwise. With a carefully decided p , the mechanism can achieve ε' -DP ($\varepsilon_2 < \varepsilon' < \varepsilon_1$).

Let $\mathcal{M} := \mathcal{S}_2 \circ (\mathcal{R}_1, \mathcal{R}_2)$ be the new mechanism. According to previous analysis,

$$\frac{\Pr[\mathcal{M}(t_i) = pt_i]}{\Pr[\mathcal{M}(t'_i) = pt_i]} \leq \frac{p \cdot e^{\varepsilon_1} q_1 + (1-p) \cdot e^{\varepsilon_2} q_2}{p \cdot q_1 + (1-p) \cdot q_2}. \quad (2)$$

And $e^{-\varepsilon_1/2} < q_1 < e^{\varepsilon_1/2}$, $e^{-\varepsilon_2/2} < q_2 < e^{\varepsilon_2/2}$.

Solving (2) $\leq e^{\varepsilon'}$, we obtain

$$p \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}{(e^{\varepsilon_1} - e^{\varepsilon'}) \cdot q_1 + (e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}$$

In Section 3.2, $\varepsilon_2 = 0$ and $q_2 = 1$. The two inequalities match.

Since $\frac{q_1}{q_2} > e^{-(\varepsilon_1 + \varepsilon_2)/2}$, then

$$p \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2})}{(e^{\varepsilon_1} - e^{\varepsilon'}) \cdot e^{-(\varepsilon_1 + \varepsilon_2)/2} + (e^{\varepsilon'} - e^{\varepsilon_2})}.$$

By setting p accordingly, the new mechanism satisfies ε' -DP.

Remarks. Some interesting results of ε' and p :

1. $\varepsilon' = \varepsilon_2$, then $p = 0$;
2. $\varepsilon' = \varepsilon_1$, then $p = 1$;
3. $\varepsilon' = (\varepsilon_1 + \varepsilon_2)/2$, then $p = \frac{1}{1 + e^{-\varepsilon_2}}$.
4. (Result calculated by GPT) $p = 1/2$, then $\varepsilon' = \ln \left(\frac{e^{(\varepsilon_1 - \varepsilon_2)/2} + e^{\varepsilon_2}}{1 + e^{-(\varepsilon_1 + \varepsilon_2)/2}} \right)$.

Special case $\varepsilon_1 = \infty$. When $\varepsilon_1 = \infty$, the mechanism directly outputs the input without perturbation. It provides no protection. We consider the inequality (2):

- If $pt_i = t_i$, (2) = $\frac{p|\mathcal{V}| + (1-p)e^{\varepsilon_2}q_2}{0 + (1-p)q_2} \leq e^{\varepsilon'}$. Thus $p \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}{|\mathcal{V}| + (e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2} \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2}) \cdot e^{\varepsilon_2/2}}{|\mathcal{V}| + (e^{\varepsilon'} - e^{\varepsilon_2}) \cdot e^{\varepsilon_2/2}}$;
- If $pt_i = t'_i$, (2) $\leq \frac{0 + (1-p)e^{\varepsilon_2}q_2}{p|\mathcal{V}| + (1-p)q_2} \leq 1 \leq e^{\varepsilon'}$;
- Otherwise, (2) $\leq \frac{0 + (1-p)e^{\varepsilon_2}q_2}{0 + (1-p)q_2} = e^{\varepsilon_2} \leq e^{\varepsilon'}$.

3.3.1 Detailed Analysis

Let $\mathcal{M} := \mathcal{S}_2 \circ (\mathcal{R}_1, \mathcal{R}_2)$ be the new mechanism. According to previous analysis, for any neighbouring instances $t_i \sim t'_i$ and any output pt_i , we should have

$$\frac{\Pr[\mathcal{M}(t_i) = pt_i]}{\Pr[\mathcal{M}(t'_i) = pt_i]} = \frac{p \cdot \Pr[\mathcal{R}_1(t_i) = pt_i] + (1-p) \cdot \Pr[\mathcal{R}_2(t_i) = pt_i]}{p \cdot \Pr[\mathcal{R}_1(t'_i) = pt_i] + (1-p) \cdot \Pr[\mathcal{R}_2(t'_i) = pt_i]} \leq e^{\varepsilon'}. \quad (3)$$

$$\frac{p \cdot \exp(-\varepsilon_1/2 \cdot d(t_i, pt_i)) \cdot C_{1,t_i} + (1-p) \cdot \exp(-\varepsilon_2/2 \cdot d(t_i, pt_i)) \cdot C_{2,t_i}}{p \cdot \exp(-\varepsilon_1/2 \cdot d(t'_i, pt_i)) \cdot C_{1,t'_i} + (1-p) \cdot \exp(-\varepsilon_2/2 \cdot d(t'_i, pt_i)) \cdot C_{2,t'_i}} \leq e^{\varepsilon'}.$$

3.4 Generalization to GP

GP takes $\Pr[\mathcal{R}(x) = y] \propto e^{-\varepsilon d(x,r)/2}$ and satisfies

$$\frac{\Pr[\mathcal{R}(x) = y]}{\Pr[\mathcal{R}(x') = y]} \leq e^{\varepsilon \cdot d(x, x')}$$

Let $d = d(x, x')$ then the SA wants to get

$$\frac{\Pr[\mathcal{M}(x) = y]}{\Pr[\mathcal{M}(x') = y]} = \frac{p \cdot e^{\varepsilon_1 d} q_1 + (1-p) \cdot e^{\varepsilon_2 d} q_2}{p \cdot q_1 + (1-p) \cdot q_2} \leq e^{\varepsilon' d}$$

We can derive the same result as,

$$p \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}{(e^{\varepsilon_1} - e^{\varepsilon'}) \cdot q_1 + (e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}$$

But we do not have the same inequality for q_1 and q_2 . Only $e^{-\varepsilon_1 d_{\max}/2} < q_1 < e^{-\varepsilon_2 d_{\max}/2}$. And $q_1/q_2 > e^{-(\varepsilon_1 + \varepsilon_2) d_{\max}/2}$.

Can we calculate $q_{1_{\max}}$ and $q_{2_{\min}}$? Compute it through the matrix?

4 Our System

Algorithm 1: DP Perturbed Prompts Warehouse

Protocol: Offline Prepare

Input: Plaintext prompt $x = (t_1, t_2, \dots, t_n)$, privacy parameters $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k)$, vocabulary $\mathcal{V}$

Output: Perturbed prompts $(\mathcal{X}^0, \mathcal{X}^1, \dots, \mathcal{X}^{k+1})$

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1  $\mathcal{X}^0 \leftarrow (\text{Rand}(\mathcal{V}))^n;$  //  $\varepsilon_0 = 0$ 
2 for  $i \leftarrow 1$  to  $k$  do
3    $\mathcal{X}^i \leftarrow \text{BaseDPPerturb}(x, \varepsilon_i);$ 
4  $\mathcal{X}^{k+1} \leftarrow x;$  //  $\varepsilon_{k+1} = \infty$ 
5 return  $\mathcal{X}$ 

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Protocol: Online Query

Input: Privacy parameter ε

Output: Perturbed prompt

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6  $i \leftarrow$  the ID such that  $\varepsilon_i \leq \varepsilon < \varepsilon_{i+1};$ 
7  $(a_1, a_2, \dots, a_n), (b_1, b_2, \dots, b_n) \leftarrow \mathcal{X}^i, \mathcal{X}^{i+1};$ 
8 if  $i = k$  then
9    $p \leftarrow 1 - \frac{1}{(e^\varepsilon - e^{\varepsilon_i}) \cdot e^{\varepsilon_i/2}};$ 
10 else
11    $p \leftarrow \frac{e^\varepsilon - e^{\varepsilon_i}}{(e^{\varepsilon_{i+1}} - e^\varepsilon) \cdot e^{-(\varepsilon_i + \varepsilon_{i+1})/2} + (e^\varepsilon - e^{\varepsilon_i})};$ 
12 for  $j \leftarrow 1$  to  $n$  do
13    $y \leftarrow \text{Ber}(p);$ 
14    $t_j \leftarrow a_j$  if  $y = 1$  else  $b_j;$ 
15 return  $(t_1, t_2, \dots, t_n)$ 

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In our DP perturbed prompts warehouse, we perturb each prompt x with some pre-defined privacy budgets $\varepsilon_1 < \varepsilon_2 < \dots < \varepsilon_k$, and store them $\mathcal{X}^i = \mathcal{R}(x, \varepsilon_i)$ in the warehouse. We all add two special case: $\varepsilon_0 = 0$ which stores $\mathcal{X}^0$ as a random choice of $\mathcal{V}$ of length $|x|$, and $\varepsilon = \infty$ which stores the original

Once a user asks for prompts with privacy budget ε (In traditional DP warehouse [?], this is usually related to the money she pays for.), we search the warehouse to find $\varepsilon_l, \varepsilon_r$ such that $\varepsilon_l \leq \varepsilon \leq \varepsilon_r$. Then we calculate p and use our sampling mechanism to output a new perturbed prompt satisfying ε -DP.

In this case, we only need to perform n Bernoulli samples. It is much more efficient than re-compute a new perturbed prompt with ε , but can (may) reach the same accuracy as re-compute ones; (must) have better accuracy than the fixed budget method.

Quick response time and light computation workload, no embedding model needed.

High accuracy level under the same privacy guarantee.

4.1 Secure Version

It can be implemented in “secure outsourced computation” framework. Generally speaking, we encrypt the warehouse and compute the sampling mechanism under MPC/TEE. Since ε is public information, the whole computation procedure is also MPC/TEE-friendly.

References

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